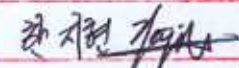


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부장 (APPROVED BY)	명칭 (NAME) ELECTRIC MOTOR DRIVEN IN-TANK FUEL PUMP FOR AUTOMOBILES				

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1. Scope

This specification defines the requirements of an electric motor driven gasoline pump (hereinafter called "Pump") located in the vehicle fuel tank.

They shall be divided into types as follows:

Rated Voltage	Types	Delivery Pressure-Flow Rate (Reference)
12 V	A	2.5 kgf/cm ² - 90 l/hr
	B	0.7 kgf/cm ² - 66 l/hr
	C	0.2 kgf/cm ² - 50 l/hr
	D	0.1 kgf/cm ² - 25 l/hr
	E	3.35 kgf/cm ² - 75 l/hr
	F	3.35 kgf/cm ² - 90 l/hr

2. Reference Specifications

This specification is based on the following JIS standards and obtained by partial addition and modification.

JIS D 3606 : Electric Fuel Pumps for Automobiles

JIS K 2202 : Motor Gasoline

JIS D 1601 : Vibration Testing Method for Automobile Parts

JIS D 5005 : Testing Voltages for Automobile Electric Equipments

JIS C 0904 : Testing Methods for Electrical Apparatus for Use in Explosive Gas Atmospheres

3. Construction

3.1 Shape and Dimension

They shall be specified on the drawing.

3.2 Fireproof Construction

(1) Pumps shall have a construction that allows no fire, which broke out even inside the pump, to run away to

outside (inside tank).

- (2) Pump outside shall have no construction that will possibly cause fire. (Electrical wirings, etc. shall have a secure fire-proof construction.)

4. Performance

- 4.1 Test method shall conform to JIS D 3606.
- 4.2 Applicable fuel shall conform to JIS K 2202.
- 4.3 Terminal Voltage : Unless stated, 12V.
- 4.4 The following items shall conform to specifications on the drawing.
- (1) Shut-off pressure
 - (2) Pump deliver
 - (3) Minimum operating voltage
 - (4) Maximum operating current
 - (5) Pressure pulse
 - (6) Noise

5. Quality

- 5.1 Fuel Leakage from Pump
- No leakage that might affect the performance in section 4.
- 5.2 Insulation
- No open, short of wires, and other defects.
- 5.3 Water Resistance
- No defects even after 6.1 Pump Alone Durability Test.
- 5.4 Corrosion Resistance
- No rust forming and other defects in all components of pumps even after 6.1 Pump Alone Durability Test.
- 5.5 Nipple Strength
- (1) Apply a load 3.5 kgf max, in the direction perpendicular to the axis, to top of the nipple. No

crack, permanent set, and other defects.

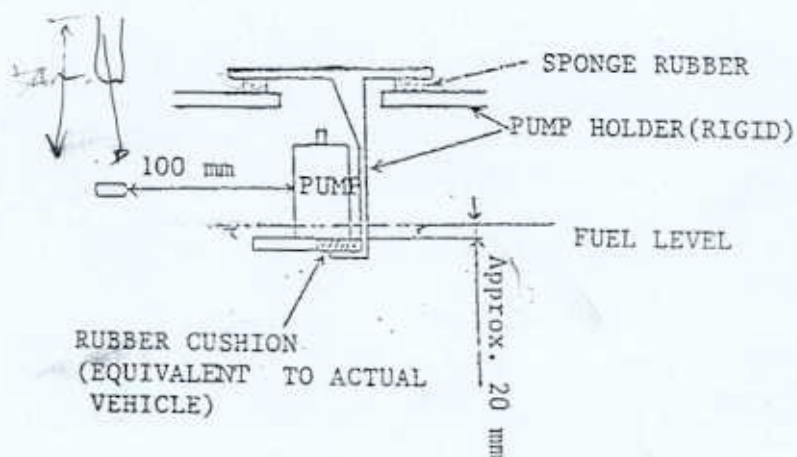
- (2) For separate nipple pumps, besides above (1), the installing location shall be free from crack and other defects when installed by regular practice.

5.6 Temperature Guarantee

Pumps shall function correctly at - 30 to 80°C.

5.7 Sound Level

Pumps shall conform to the sound level specified on the drawing during operation at room temperature and rated voltage in the installation state as shown below or equivalent.



5.8 Radio Wave Interference

No objectionable noise or interference to the radio and other equipments that are normally expected to be mounted on the vehicle.

Note : Defects evaluation and remedy shall be arranged under agreement between the vendor and HMC.

5.9 Explosion Prevention

Conduct explosive firing test in accordance with JIS C 0904. No fire shall run away to outside the pump (inside the tank). After test, no excessive change in various functional parts.

Test conditions shall be as follows:

Explosion class : Class 1

Applicable gas :

Mixture of hydrogen 57 to 60 % (by volume) and air,
the remainder.

Test repetition : 15 cycles.

Miscellaneous : Rubber hoses and filters shall be
removed before test.

5.10 Vibration Resistance(Pump Body Alone)

Conduct resonance vibration test and vibration durability
test in accordance with Type 1, Class B in JIS D 1601.

No defects in function.

5.11 Durability

After endurance test in accordance with section 6, pump
shall have at least 90 % of the performance in section 4
and free from excessive change in the quality of section 5
(5.11 and 5.12 excluded) and in other functions that might
affect serviceability.

5.12 Material

The composing material of pumps shall be adequate to the
applicable location as shown below.

(1) Metallic Material

In the service condition the material shall be
corrosion resistant for itself or given rust preventive
treatment. It shall have sufficient resistance to
impact and friction.

(2) Nonmetallic Material

The nonmetallic material shall not corrode metallic
materials in contact and shall produce no electrical

insulators. The material itself shall have a sufficient resistant to fuel[see 6.1, (3) Fuel and durability.

6. Endurance Test

6.1 Pump Alone Durability

Use different pumps as test specimen per individual test items in Table 1. Conduct test in accordance with the conditions given.

Notes :

(1) Install a fuel filter, equivalent to regular service part, on the pump outlet.

(2) The fuel and the fuel filter on outlet shall be maintained so as not to change excessively as compared with the initial condition.

(3) Use the following fuels :

(a) Fuel D : Mixture of isooctane 60 parts and toluene 40 parts by volume.

(b) Unleaded gasoline:

Commercial unleaded gasoline
(0.5 % water included)

(c) High-octane gasoline:

Leaded gasoline with lead content approx. 0.7g/l.

(d) Acidified gasoline:

Unleaded gasoline including LPO 1 % by weight
(replaced with fresh gasoline every 50 hr in both room-and high-temperature run).

(e) Alcohol gasoline :

Unleaded gasoline including normal industrial methanol 15 % by volume(check periodically for

excessive change in alcohol concentration,
or replace).

(f) Water-included gasoline :

Unleaded gasoline including water 3% by volume.

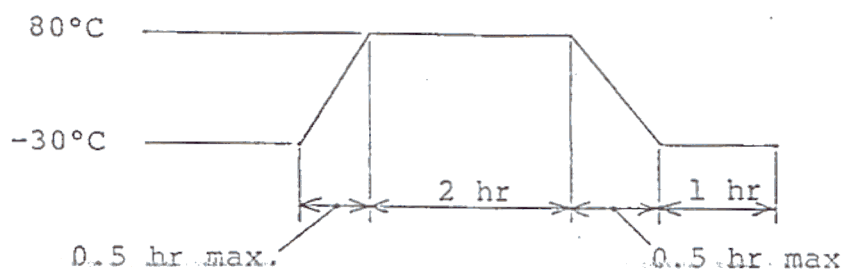
(g) Premium gasoline :

Commercial unleaded premium gasoline.

6.2 Pump Alone Heat Cycle Test

Run 20 cycles of the following test on the pump in dry state.

One cycle shall consist of :



One cycle

6.3 Dry Run Test

Open the delivery outlet of the pump from wet state and run without fuel supply at 14V for continuous 20 minutes.

6.4 Overvoltage Run Test

Run at 18V and fuel temperature 40°C with the delivery outlet under the following conditions for continuous 24 hr.

Type A,E,F : delivery pressure 3.5 kgf/cm².

Type B : delivery pressure 0.7 kgf/cm².

Type C, D : delivery outlet open

6.5 Durability Test by Vehicle

Various vehicle tests durability shall be conducted in accordance with MMC durability run test standard.

Note : The test, as a rule, shall be performed in the course of new vehicle development by MMC.

TABLE 1. PUMP ALONE DURABILITY RUN TEST

PERFORM MARK ○

Test Item	Test Specimen (Including $\frac{1}{2}$ " Filter)		Test Fuel						Temperature °C (Fuel and Pump)	Test Conditions		
	New Product	Fuel D-Aged *	Fuel D	Unleaded Gasoline	High-Octane Gasoline	Acidified Gasoline	Alcohol Gasoline	Water-Included Gasoline	Premium Gasoline	Voltage (V)	Delivery Pressure	Test Time (hr)
Room Temperature Condition Run	○	○	○	○				○		13.5 ± 0.5	See Table below (For Types A, E, F and B, see **)	Run : 0.2 Standing : 168
Room and Standing	○	○	○	○								300
	○											300
	○				○							300
	○					○						200
	○								○			300
High Temperature Continuous Run	○									10 ± 0.5		500
	○								40			
	○								-30			100

Note * : Soaked in Fuel D at 60°C for 100 hr.

TABLE 1 (continued)

** Run time divisions for types A,E,F and B are as follows.

Type	Delivery Pressure or Jet Dia	Test Time (hr)
A	2.5 kgf/cm ²	3,000
	3.5 kgf/cm ²	1,000
	Total	4,000
B	0.2 kgf/cm ²	1,000
	0.7 kgf/cm ²	3,000
	Total	4,000
C	ø 2	4,000
D	ø 1.7	4,000
E, F	3.3 kgf/cm	3,000
	4.3 kgf/cm	1,000
	Total	4,000

Type	Delivery Pressure (kgf/cm ²)	Time Ratio
A	2.5	3
	3.5	1
B	0.2	1
	0.7	3
E, F	4.3	1
	3.3	3

7. Service Conditions

Pump shall permit the use for 8 years or 120,000 km without replacement under the conditions below :

- (1) Installing Location : Inside fuel tank
- (2) Installing Position : Inlet facing downward
- (3) Ambient Temperature : -30 to 80°C
- (4) Applicable Fuel :

Sometimes alcohol-included fuel is used, and sometimes water, dust, etc. are contained.

- (5) Application :

Passenger cars and trucks for Korea , North America, Europe, and other countries.

- 8. Vendor drawings shall be submitted and treated in accordance with Purchase Parts Drawing Handling Procedure.